The Hong Kong University of Science and Technology
Department of Computer Science and Engineering
COMP4421 (Fall 2022)

Midterm Examination

Name:

Student ID:

# Overview

Time: 9:00 am – 10:00 am, Oct 18, 2022

Scores:

| Question | Maximum Score | Your Score |
|---|---|---|
| 1 | 20 | |
| 2 | 20 | |
| 3 | 20 | |
| 4 | 20 | |
| 5 | 20 | |
| Total | 100 | |

# Question 1

Consider a 3-bit image of size M×N=50×40, and the intensity distribution is shown in the following table, where $r_k$ denotes intensity value ($r_k = 0, \dots, L-1$), $n_k$ denotes the number of pixels in the image with intensity value $r_k$.

| $r_k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $n_k$ | 300 | 240 | 160 | 400 | 100 | 250 | 200 | 350 |

Table 1.1 Original image

(1) Fill out the following tables with calculations for histogram equalization. CDF denotes the cumulative distribution function.

| $r_k$ | $n_k$ | $p_r(r_k) = n_k/MN$ | CDF | CDF*(L-1) | Round off |
|---|---|---|---|---|---|
| 0 | | | | | |
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |

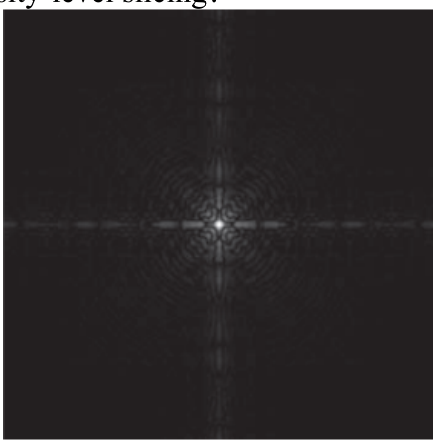
Table 1.2 Histogram equalization

| $r_k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $n_k$ | | | | | | | | |

Table 1.3 Equalized image

(2) Histogram equalization could enhance the contrast of images, is histogram equalization always good? Please justify your answer.

(3) Consider the following images, which of the following transformation (or transformations) is proper to enhance the following images: (i) log transformation, (ii) power-law transformation with $\gamma < 1$, (iii) power-law transformation with $\gamma > 1$ (iv) intensity-level slicing?



(a)

(b)

Solution:

(1)

Table 1.2 Histogram Equalization:

| $r_k$ | $n_k$ | $p_r(r_k) = n_k/MN$ | CDF | CDF*(L-1) | Round off |
|---|---|---|---|---|---|
| 0 | 300 | 0.15 | 0.15 | 1.05 | 1 |
| 1 | 240 | 0.12 | 0.27 | 1.89 | 2 |
| 2 | 160 | 0.08 | 0.35 | 2.45 | 2 |
| 3 | 400 | 0.2 | 0.55 | 3.85 | 4 |
| 4 | 100 | 0.05 | 0.6 | 4.2 | 4 |
| 5 | 250 | 0.125 | 0.725 | 5.075 | 5 |
| 6 | 200 | 0.1 | 0.825 | 5.775 | 6 |
| 7 | 350 | 0.175 | 1.00 | 7 | 7 |

Table 1.3 Equalized image:

| $r_k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $n_k$ | 0 | 300 | 400 | 0 | 500 | 250 | 200 | 350 |

(2)

No. It may increase the contrast of background noise, while decreasing the useful signal or it can remove small details.

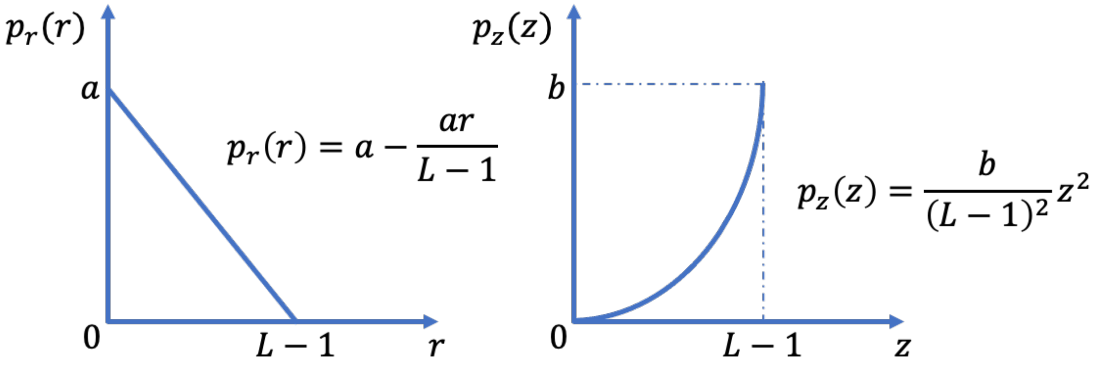
(3)

For figure (a), we should apply log transformation or power-law transformation with $\gamma < 1$

For figure (b), we should apply power-law transformation with $\gamma > 1$.

# Question 2

An image with intensities in the range $[0, L-1]$ has the probability density function $p_r(r)$ shown in Figure (a). It is desired to transform the intensity levels of this image so that they will have the specified probability density function $p_z(z)$ as shown in Figure (b). Assume continuous quantities and find the transformation $z = f(r)$ that will accomplish this.




(a) (b)

Solution:

For figure (a), we have the histogram equalization equation as:

$$s = T(r) = (L-1)\int_0^r p_r(w)\,dw$$

After calculation we get:

$$s = T(r) = (L-1)\int_0^r a - \frac{aw}{L-1}dw = (L-1)ar - \frac{a}{2}r^2$$

For figure (b), we do the same calculation:

$$s = G(z) = (L-1)\int_0^z p_z(w)\,dw = (L-1)\int_0^z \frac{b}{(L-1)^2}w^2 dw = \frac{b}{3(L-1)}z^3$$

Then we could get the relationship between $z$ and $s$:

$$\Rightarrow z = G^{-1}(z) = \sqrt[3]{\frac{3(L-1)s}{b}}$$

Substitute equation of $s$ into equation of $z$, we get

$$z = \sqrt[3]{\frac{3(L-1)s}{b}} = \sqrt[3]{\frac{3(L-1)((L-1)ar - \frac{a}{2}r^2)}{b}}$$

# Question 3

Consider the following image $f$ in Table 3.1 and filter $h$ in Table 3.2:

| x \ y | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 2 | 3 | 4 |
| 2 | 2 | 4 | 6 | 8 |
| 3 | 3 | 6 | 9 | 12 |
| 4 | 4 | 8 | 12 | 16 |

Table 3.1

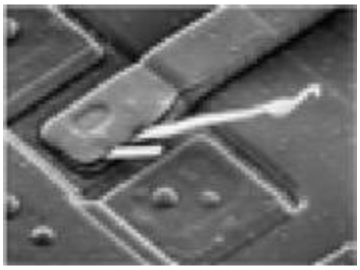
| x \ y | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 1 | 2 | 1 |
| 2 | 0 | 0 | 0 |
| 3 | -1 | -2 | -1 |

Table 3.2

(1) When the filter centers at the position (x=2, y=2), compute the correlation and convolution result, respectively.
(2) When applying the filter $h$ to the image $f$, describe the effects of image enhancement.
(3) Please give the results of performing a median filter on $f$ using a $3 \times 3$ filter. You only need to compute response for pixels where the entire filter overlaps with the image (that is, the result image should have the shape size $2 \times 2$).
(4) When applying the filter $h$ to image $f$, consider the convolution computation $h \star f$, do you have an efficient way to compute the result?

(Hint: $h$ is separable, i.e., $h = h_1 h_2^\intercal$, where $h_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and $h_2^\intercal = [1,2,1]$.)

(5) Consider the following images:


(a)


(b)


(c)

Figure (a) is the original image. Figure (b) and (c) are the results of iteratively applying 5 times filtering using median filter and box filter, respectively. What if we iteratively apply smooth filtering with median filter on Figure (b) and box filter Figure (c), separately? How would Figure (b) and (c) look like?

Solution:

a) the correlation result:

$$h(1{,}1)f(1{,}1) + h(1{,}2)f(1{,}2) + h(1{,}3)f(1{,}3)$$
$$+h(2{,}1)f(2{,}1) + h(2{,}2)f(2{,}2) + h(2{,}3)f(2{,}3)$$
$$+h(3{,}1)f(3{,}1) + h(3{,}2)f(3{,}2) + h(3{,}3)f(3{,}3)$$
$$= 1\times 1 + 2\times 2 + 3\times 1 + 2\times 0 + 4\times 0 + 6\times 0 + 3\times(-1) + 6\times(-2) + 9\times(-1)$$
$$= -16$$

For convolution, first rotate $h$ 180 degrees:

$$h_p = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

The convolution result is:

$$h_p(1{,}1)f(1{,}1) + h_p(1{,}2)f(1{,}2) + h_p(1{,}3)f(1{,}3)$$
$$+h_p(2{,}1)f(2{,}1) + h_p(2{,}2)f(2{,}2) + h_p(2{,}3)f(2{,}3)$$
$$+h_p(3{,}1)f(3{,}1) + h_p(3{,}2)f(3{,}2) + h_p(3{,}3)f(3{,}3)$$
$$= 1\times(-1) + 2\times(-2) + 3\times(-1) + 2\times 0 + 4\times 0 + 6\times 0 + 3\times 1 + 6\times 2 + 9\times 1$$
$$= 16$$

b) Image sharpening (or edge detection).

The difference between the third and first rows of the $3\times 3$ image region approximates the partial derivative in the x-direction, and is implemented using this kernel. It sharpens the edge of the image in the horizontal direction.

The idea behind using a weight value of 2 in the center coefficient is to achieve some smoothing by giving more importance to the center point.

The coefficients in this kernel sum to zero, so they would give a response of zero in areas of constant intensity, as expected of a derivative operator.

c) $\begin{bmatrix} 3 & 6 \\ 6 & 8 \end{bmatrix}$

d)

The filter $h$ has size $m \times n$ and the size of the image $f$ is $M \times N$. The total computation is in the order of $MNmn$.

The more efficient way to compute the $h \star f$ is:

As the filter $h$ is separable, so $h = h_1 h_2^{\mathrm{T}}$, where $h_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and $h_2^{\mathrm{T}} = [1{,}2{,}1]$.

By associative, $h \star f = (h_1 \star h_2) \star f = h_1 \star (h_2 \star f)$. The computation of the first convolution $g = h_2 \star f$ is in the order of $MNn$ and computation of the second convolution $h_1 \star g$ is in the order of $MNm$. The total computation is in the order of $MN(m+n)$.

So, the computation advantage is $\frac{mn}{(m+n)} = 1.5$

~~e) For median filter, the image will converge to an image invariant to the filter.~~

~~For box filter, the image will converge to an image with constant intensity.~~

# Question 4

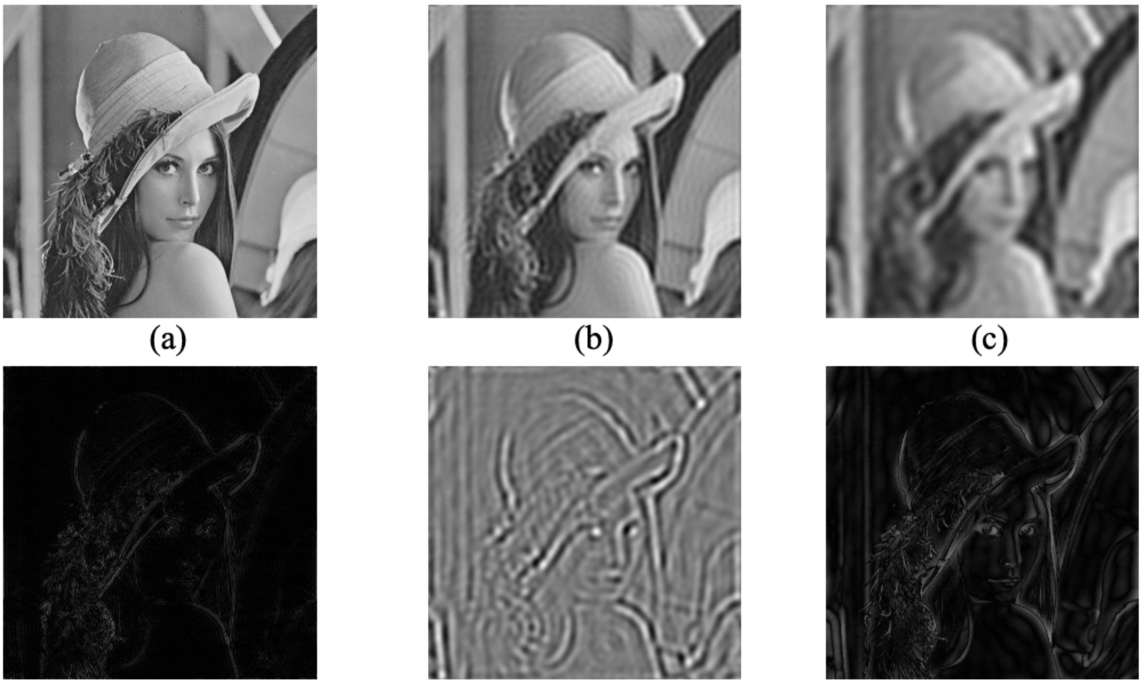
(a) (b) (c)

Figure 4.1

(1) Figure 4.1 shows three images. (a) and (b) are original images. (c) is reconstructed image by using Fourier transformation information from (a) and (b). Which phase information does image (c) use during reconstructing? _______

(i) Image (a)'s phase angle.

(ii) Image (b)'s phase angle.



(a) (b) (c)

(d) (e) (f)

Figure 4.2

(2) Figure 4.2 shows six images. Image (a) is the original image. The rest of images are the results of filtering using Ideal Lowpass Filter (ILPF), Ideal Highpass Filter (IHPF) and band-pass filter. The five filters are shown as follows ($\boldsymbol{D_0}$ is the cutoff frequency):

(i) ILPF with $\boldsymbol{D_0 = 20}$.

(ii) ILPF with $\boldsymbol{D_0 = 40}$.

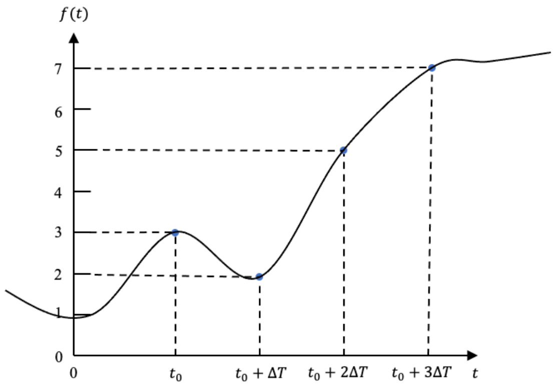
(iii) IHPF with $\boldsymbol{D_0 = 10}$.

(iv) IHPF with $\boldsymbol{D_0 = 40}$.

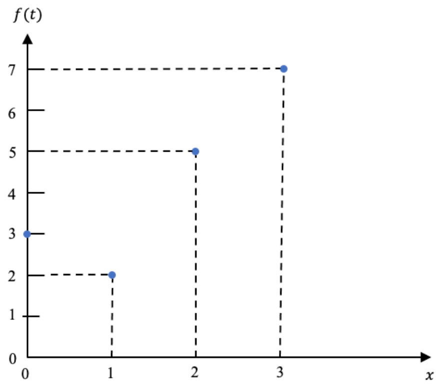
(v) Band-pass filter.

Please fill in the blanks with corresponding filters.

| Image | Filter |
|---|---|
| (b) | |
| (c) | |
| (d) | |
| (e) | |
| (f) | |




(a)




(b)

Figure 4.3

(3) Figure 4.3 (a) shows four samples of a continuous function, $f(t)$, taken $\Delta T$ units apart. Figure 4.3 (b) shows the samples in the discrete domain. The value of $x$ are 0, 1, 2, and 3, which refer to the number of samples in sequence, counting from 0. For example, $f(1) = f(t_0 + \Delta T)$. Given the discrete Fourier transformation formula $F(u) = \sum_{x=0}^{M-1} f(x)e^{-j2\pi ux/M}$ $u = 0, 1, \ldots, M-1$, please compute the values of $F(0), F(1)$.

(1) (ii). **(4pts)**

(2) **(10pts)**

| Image | Filter |
|---|---|
| (b) | (ii) |
| (c) | (i) |
| (d) | (iv) |
| (e) | (v) |
| (f) | (iii) |

**(3) (6pts)**

$f(0) = 3, f(1) = 2, f(2) = 5, f(5) = 7,$

$$F(0) = \sum_{x=0}^{3} f(x)e^{0} = f(0) + f(1) + f(2) + f(3) = 17,$$

$$F(1) = \sum_{x=0}^{3} f(x)e^{-j2\pi x/4} = 3 + 2e^{-j\pi/2} + 5^{-j\pi} + 7e^{-j3\pi/2}$$

$$= 3 + 2(cos(-\frac{\pi}{2}) + jsin(-\frac{\pi}{2})) + 5(cos(-\pi) + jsin(-\pi)) + 7(cos(-\frac{3\pi}{2}) + jsin(-\frac{3\pi}{2})),$$

$$= 3 + 2(0 - j) + 5(-1 + 0) + 7(0 + j),$$

$$= 3 - 2j - 5 + 7j = -2 + 5j,$$

# Question 5

Consider a 3-bit grayscale image $g(r, c)$ of size $5 \times 5$ as shown in Table 5, you need to apply some filters on the image to get the filtered image $f(x, y)$.

| x \ y | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 0 | 2 | 1 | 4 | 5 |
| 2 | 7 | 4 | 5 | 3 | 6 |
| 3 | 1 | 0 | 3 | 7 | 3 |
| 4 | 2 | 3 | 1 | 0 | 2 |
| 5 | 1 | 1 | 2 | 5 | 5 |

Table 5

(1) Please calculate the result of applying a $3 \times 3$ Midpoint Filter centered at (x=3, y=2).

$$\text{Midpoint Filter: } f(x,y) = \frac{1}{2}\left[\max_{(r,c)\in S_{xy}}\{g(r,c)\} + \min_{(r,c)\in S_{xy}}\{g(r,c)\}\right]$$

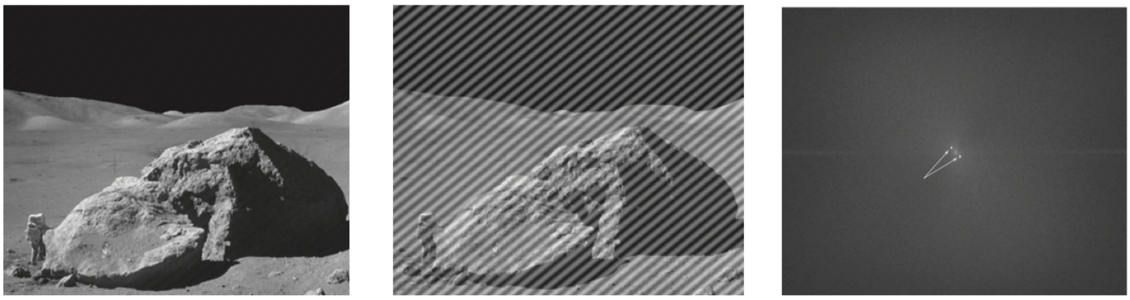
(2) Please calculate the result of applying a $3 \times 3$ Geometric Mean Filter centered at (x=2, y=4). Consider an image is corrupted by the pepper noise, can we use the Geometric Mean Filter to eliminate this type of noise? Please also explain the reason.

$$\text{Geometric Mean Filter: } f(x,y) = \left[\prod_{(r,c)\in S_{xy}} g(r,c)\right]^{\frac{1}{mn}}$$

(3) Please calculate the result of applying a $\mathbf{3 \times 3}$ Contraharmonic Mean Filter with $\boldsymbol{Q = 2}$ centered at (x=4, y=3). Consider an image is corrupted by salt-and-pepper noise, can we use this filter to eliminate the noise? Please also explain the reason.

$$\text{Contraharmonic Mean Filter: } \boldsymbol{f(x,y)} = \frac{\sum_{(r,c)\in S_{xy}} \boldsymbol{g(r,c)^{Q+1}}}{\sum_{(r,c)\in S_{xy}} \boldsymbol{g(r,c)^{Q}}}$$

(4) Figure 5 (a) is an original image. Figure 5 (b) is the corrupted image by periodic noise from (a). Figure (c) is the Fourier spectrum obtained by applying Fourier transformation from (b). The two arrows in Figure 5 (c) indicate two conjugate impulses. Please state how to restore the original image from Figure 5 (b)?



(a) (b) (c)

Figure 5

**(1) (4pts)**

$\boldsymbol{f = \frac{7+0}{2} = 3.5}$

**(2) (6pts)**

$\boldsymbol{f = (1 \times 4 \times 5 \times 5 \times 3 \times 6 \times 3 \times 7 \times 3)^{1/9} = 3.644}$

Using geometric mean filter can't remove the pepper noise. The value of pepper noise is 0, so the result after multiplication is still 0 which is also pepper noise.

**(3) (6pts)**

$\boldsymbol{f = (27 + 343 + 27 + 1 + 1 + 8 + 125)/(9 + 49 + 9 + 1 + 1 + 4 + 25) = 5.43}$

Using contraharmonic mean filter with Q=2 can't remove the salt noise. The value of salt noise is extremely large which could overpower the result after processing.

**(4) (4pts)**

(i) We do the Fourier transformation and get the Fourier spectrum.

(ii) We find that the Fourier spectrum has a pair of bright dots (i.e., impulses) as shown in Figure 5 (c).

(iii) Determine the location of the dots and apply notch reject filter transfer function whose notches coincide with the location of the dots:

Notch Reject Filter: $\boldsymbol{H_{NR}(u,v) = \prod_{k=1}^{Q} H_k(u,v) H_{-k}(u,v)}$,

where the centers of these two dots are $(\boldsymbol{u_k}, \boldsymbol{v_k})$ and $(\boldsymbol{u_{-k}}, \boldsymbol{v_{-k}})$ respectively.

For example, the Butterworth Notch Reject Filter transfer function of order $\boldsymbol{n}$ with 1 notch pair is shown as follow:

$$\boldsymbol{H_{NR}(u,v) = \left[\frac{1}{1 + [D_{01}/D_k(u,v)]^n}\right]\left[\frac{1}{1 + [D_{01}/D_{-k}(u,v)]^n}\right]}$$

and,

$$\boldsymbol{D_k(u,v) = \left[(u - \frac{M}{2} - u_k)^2\right]^{1/2} - \left[(v - \frac{N}{2} - v_k)^2\right]^{1/2}}$$

$$\boldsymbol{D_{-k}(u,v) = \left[(u - \frac{M}{2} + u_k)^2\right]^{1/2} + \left[(v - \frac{N}{2} + v_k)^2\right]^{1/2}}$$

where $\boldsymbol{D_{01}}$ denotes the cutoff frequency. M and N are the size of the 2-D grayscale image.

(iv) Combine the Fourier spectrum after applying the notch reject filter and the original phase angle and apply the Inverse Fourier Transform, then we can restore the original image.